

Introduction to Immunologically-Mediated Diseases

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Session Objectives

1. Describe the mechanism by which each immunologically mediated diseases occurs
2. Distinguish immunologically mediated diseases based on clinical and laboratory signs and symptoms
3. Be able to differentiate organ specific from systemic immunologically mediated diseases
4. Be able to determine appropriate diagnostic tests for each immunologically mediated disease
5. Be able to select an appropriate treatment for an immunologically mediated disease based on the disease pathology
6. Compare and contrast clinical and laboratory findings of immunologically mediated diseases

Session Outline

Post-Streptococcal Glomerular Nephritis

Hashimoto's Thyroiditis

Pernicious anemia

Idiopathic thrombocytopenic purpura

Pemphigus vulgaris

Primary biliary cirrhosis

Anti-phospholipid syndrome

Goodpasture's Syndrome

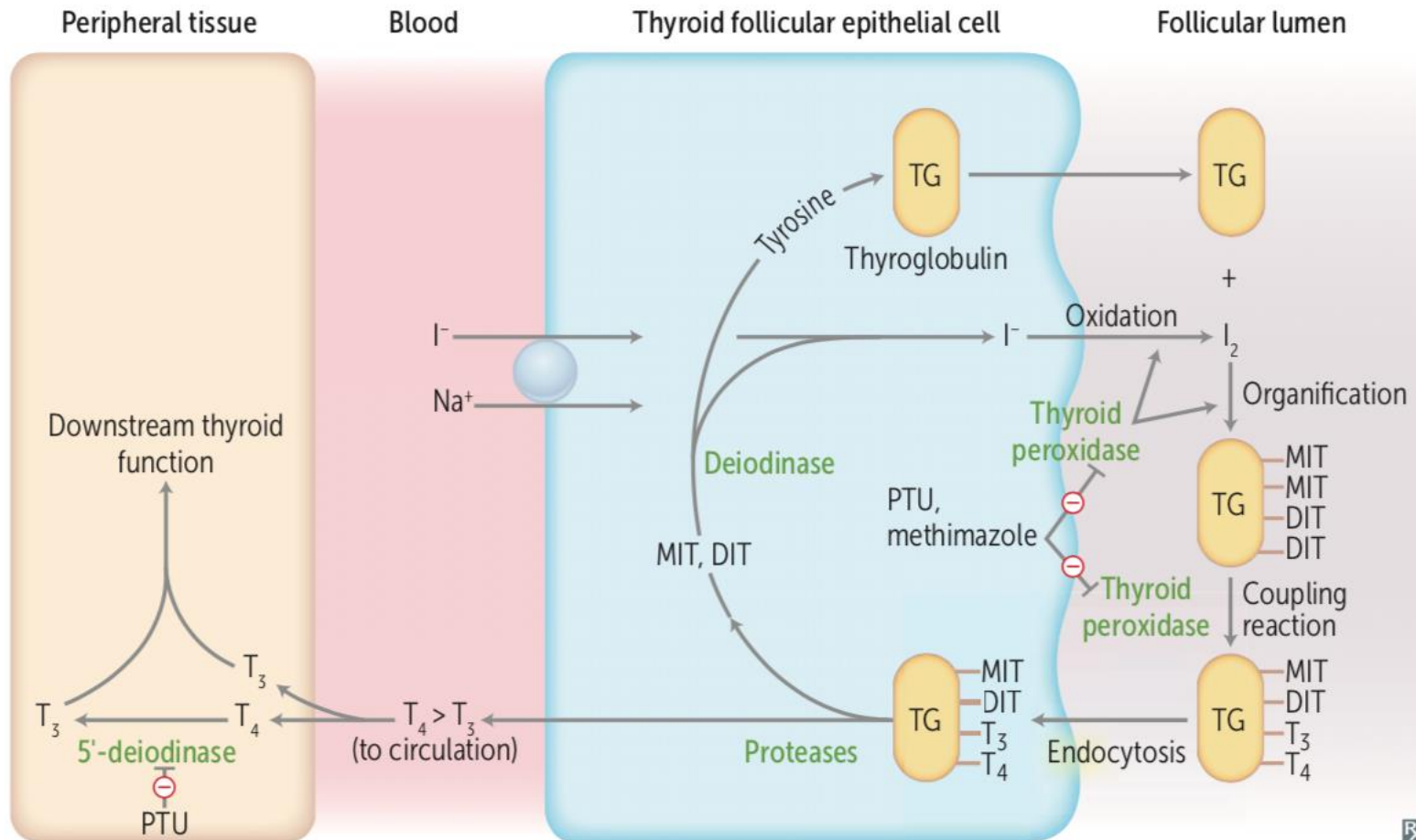
Ulcerative colitis

Insulin-dependent Diabetes Mellitus

Sjogren's syndrome

Autoimmune Diseases of Endocrine and Exocrine Glands

Hashimoto's Thyroiditis



Hashimoto's Thyroiditis

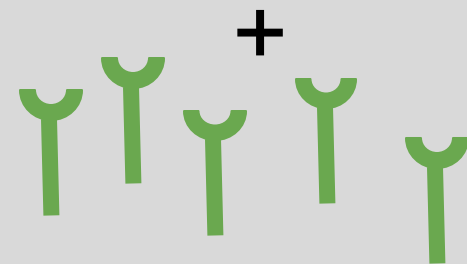
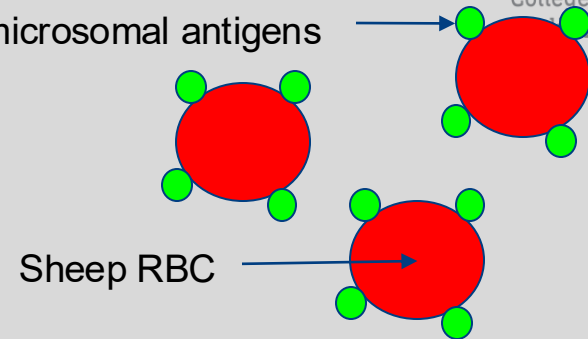
- Mechanism of Action: IgG auto-Abs against thyroid peroxidase (antimicrosomal) & thyroglobulin
- Organ-specific: Thyroid gland
- Signs & Symptoms:
 - Risk Factors: environmental (smoking, drugs), genetic (HLA-DR5 & DR3)
 - Diffuse, non-tender, moderate thyromegaly
 - Hypothyroidism

Hashimoto's Thyroiditis

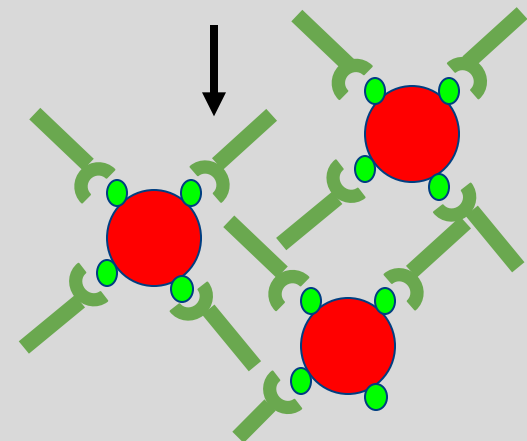
- Diagnosis:
 - Anti-TPO (microsomal) Abs in 90%(non-specific)
 - Anti-TG Abs in 40%
 - Follicle damage with chronic inflammation and Hurthle cells

- Treatment:
 - If hypothyroid, levothyroxine

Solubilized microsomal antigens



Individual's auto-Abs against microsome



Antimicrosomal Ab
hemagglutination assay

Insulin-dependent Diabetes Mellitus

- Mechanism of Action: T-cell and M ϕ infiltration of β -cells
 - Decreased insulin
- Organ specific: Pancreatic β -cells
- Signs and Symptoms:
 - Genetic associations: HLA-DR3 & DR-4 Often asymptomatic
 - DKA common initial presentation
 - Polyuria, polydipsia, glucosuria

Insulin-dependent Diabetes Mellitus

- Diagnosis:
 - Oral blood glucose testing (fasting vs. non-fasting)
 - HbA1c $\geq 6.5\%$ (monitoring)
 - Anti-glutamic acid decarboxylase auto-antibodies
 - Islet cell cytoplasmic antibodies
- Treatment:
 - Exogenous insulin
 - Pancreas transplant

Sjögren's Syndrome

- Mechanism of Action: Lymphocytic infiltration + fibrosis → Destruction; 1° or 2°
- Non-organ specific: Exocrine glands, especially lacrimal & salivary
- Signs & Symptoms:
 - Xerostomia, keratoconjunctivitis sicca, b/l parotid enlargement
 - Vasculitis, xerosis, pruritus, & annular erythema
 - Marginal zone lymphoma
 - Rheumatoid Arthritis, Hashimoto's, Primary Biliary Cholangitis, Systemic Lupus Erythematosus

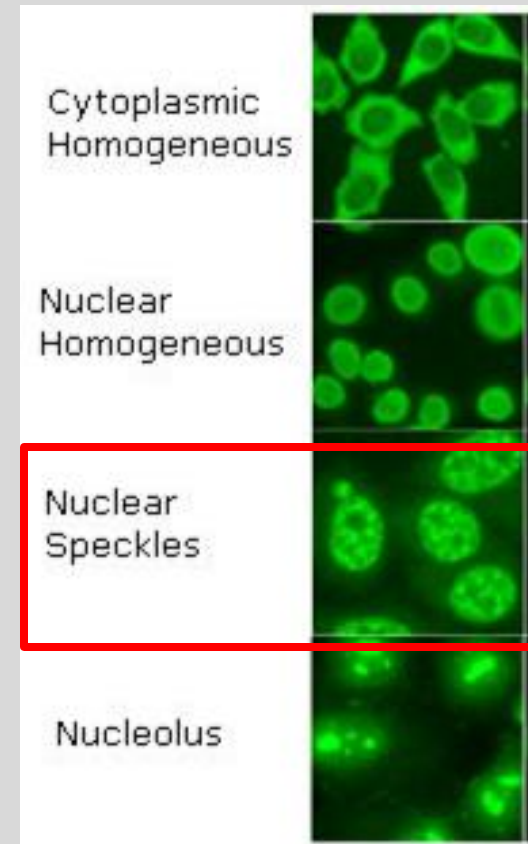
Sjögren's Syndrome

- Diagnosis:

- + RF 70%; + ANA: nuclear speckled;
Anti-Ro/SSA & Anti-La/SSB
- Schirmer test
- Biopsy

- Treatment:

- Immunosuppressive drugs (Prednisone)
- Pilocarpine, artificial tears



WoBeM:

https://wwwn.cdc.gov/Nchs/Nhanes/2003-2004/SSANA_C.htm

Case 1

A 15-year-old male presents to the ER with tachypnea, labs demonstrate elevated glucose and a high anion gap metabolic acidosis. He reports no past medical history, though family reports ‘high sugars’ in several family members at a young age. What is the underlying pathological process?

- A) Lymphocytic infiltration and fibrosis of exocrine glands
- B) Cytotoxic T cells and Macrophages to β cells
- C) Autoantibodies against IgG
- D) Lymphocytic infiltration and fibrosis of endocrine glands

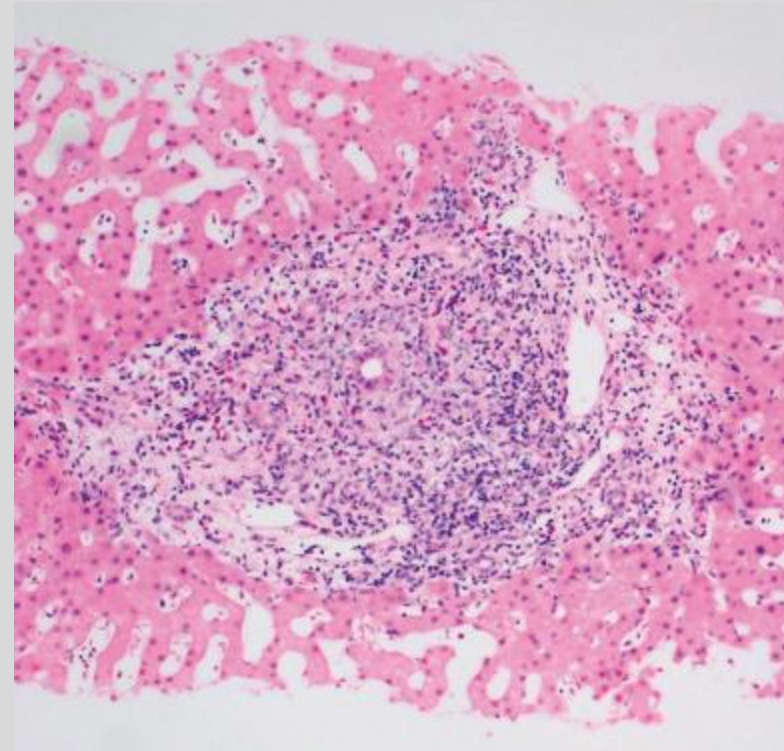
Autoimmune Diseases of the Gastrointestinal Tract

Primary Biliary Cholangitis

- Mechanism of Action: etiology unclear; lymphocytic infiltrate + granulomatous inflammation
 - Auto-Abs against mitochondria
- Organ specific: medium-sized intrahepatic bile ducts
- Signs & Symptoms:
 - Pruritus
 - Obstructive jaundice
 - Cirrhosis, portal HTN
 - Xanthomas, steatorrhea

Primary Biliary Cholangitis

- Diagnosis:
 - Obstructive liver labs
 - ↑ Alk-Phos, GGT, LFT's, hypercholesterolemia; conjugated (direct) hyperbilirubinemia
 - +AMA 93%
 - Increased IgM
 - Liver biopsy
- Treatment: Ursodeoxycholic Acid (USDA)



A portal tract in primary biliary cirrhosis contains a nodular mixed inflammatory infiltrate consisting of lymphocytes, plasma cells, numerous eosinophils, and histiocytes.

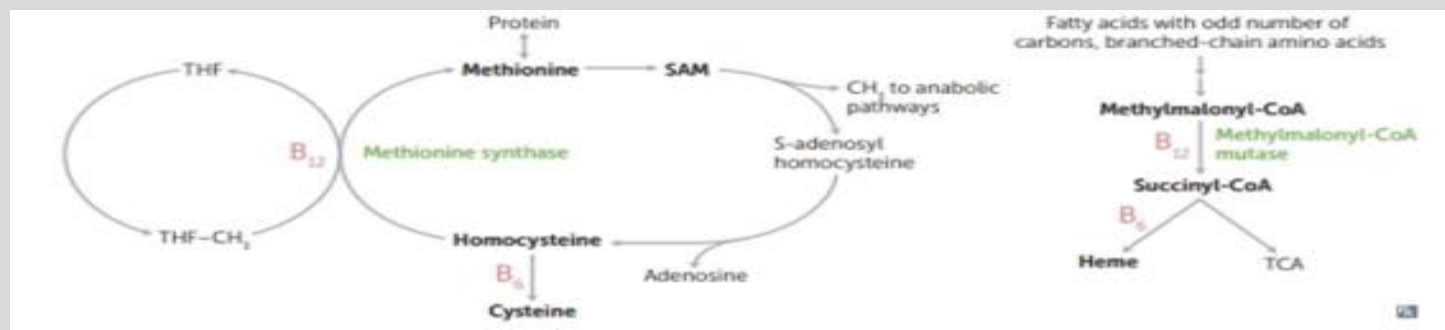
Rosai and Ackerman's Surgical Pathology, 2018

Pernicious Anemia

- Mechanism of Action: IgG auto-Antibodies against parietal cells &/or intrinsic factor
 - B12 (cobalamin) deficiency
- Organ specific: gastric parietal cells
- Signs & Symptoms:
 - Angular cheilitis, glossitis
 - Peripheral sensory neuropathy
 - Loss of proprioception
 - Pallor, weakness, fatigue

Pernicious Anemia

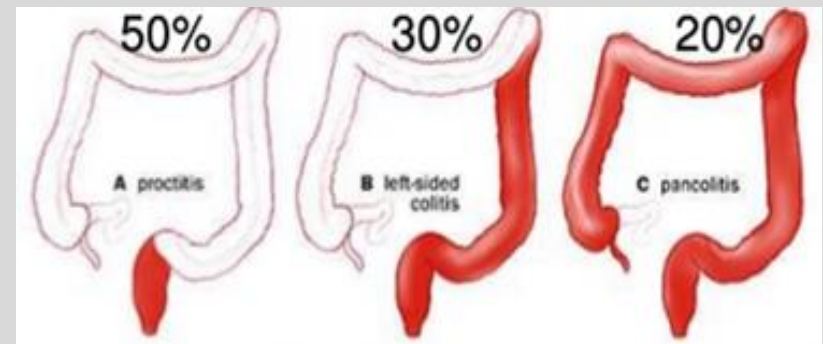
- Diagnosis:
 - Low B12, macrocytic (megaloblastic) anemia
 - Increased homocysteine and methylmalonic acid
 - Acid secretion analysis
 - Endoscopy with biopsy
 - Historically: Schilling test
- Treatment: Vitamin B12 **Intramuscular**



Rosai, J., Ackerman, L. V., Goldblum, J. R., Lamps, L. W., McKenney, J. K., & Myers, J. L. (2018). *Rosai and Ackermans surgical pathology*. Philadelphia: Elsevier.

Ulcerative Colitis

- Mechanism of Action: etiology unclear; mucosal & submucosal Th2-mediated inflammation
- Organ specific: Rectum
- Signs & Symptoms:
 - Smoking protective
 - Bloody diarrhea, Bright Red Blood Per Rectum (BRBPR), anemia, LLQ pain
 - Inflammation, oral ulcerations, pyoderma gangrenosum, 1° sclerosing cholangitis
 - Colorectal carcinoma



Ulcerative Colitis

- Diagnosis:

- Inc ESR; dec Hgb/Hct & Albumin
- +MPO/p-ANCA
- Stool bacterial culture
- Barium enema: Lead pipe sign
- Endoscopic biopsy



Double contrast barium enema
<https://radiopaedia.org/articles/lead-pipe-sign-colon?lang=us>

- Treatment:

- 5-Aminosalicylic acid preparations, sulfasalazine, 6-Mercaptopurine, infliximab
- Colectomy

Case 2

A 22-year-old female presents to the clinic with concerns due to bloody diarrhea and what she describes as ‘cold sores’ at the corners of her lips that have not healed. She reports a recent camping trip but denies any fevers or chills. Labs are as follows: What is the potential disease process at hand?:

Lab tests	Patient's Results	Normal Lab Values
Hemoglobin	10.0	12.0-18.0 g/dL
Hematocrit	31%	37-47%
Leukocyte count	8,000	4,500-11,000 mm
Erythrocyte Sedimentation Rate	50	0-20 mm in one hour
c-ANCA	-	-
p-ANCA	+	-

Autoimmune Diseases of the Kidney

Poststreptococcal Glomerulonephritis

- Mechanism of Action: IgG, IgM, & C3 deposition on glomerular basement membrane (GBM) & mesangium
 - **Molecular mimicry**: Nephritogenic group A *Streptococci* pharyngitis or impetigo
- Organ specific: Renal GBM & mesangium
- Signs & Symptoms:
 - Ranges from asymptomatic to nephrotic syndrome
 - Children
 - Tea or Coca-cola colored urine
 - Periorbital & peripheral edema
 - HTN

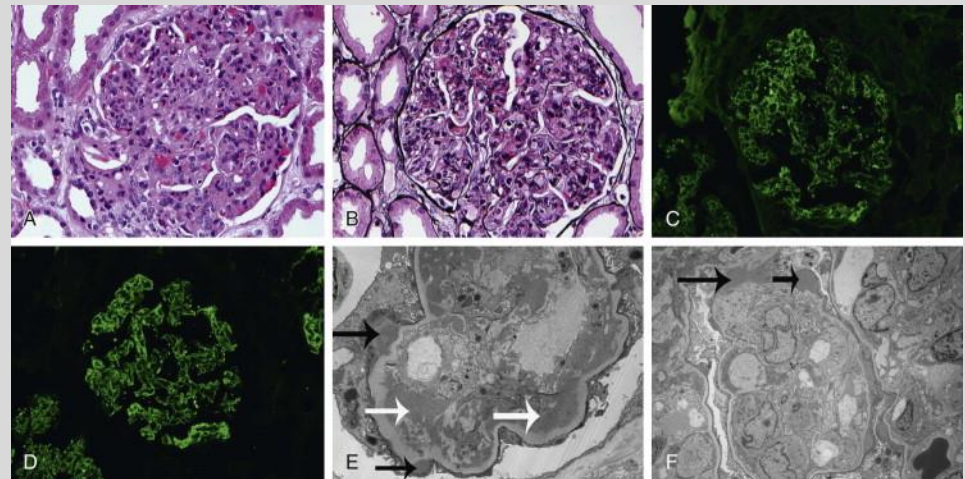
Poststreptococcal Glomerulonephritis

- Diagnosis:

- Low C3 and/or CH50
- ASO titers
- Immunofluorescence microscopy: Granular

- Treatment:

- Anti-hypertensives
- Antibiotics
- Diuretics/Na⁺ restriction
- Supportive



Poststreptococcal Glomerulonephritis

Sethi et al., Glomerular Disease. Andreoli and Carpenter's Cecil Essentials of Medicine. Published January 1, 2016. Pages 314-328

Goodpasture's Syndrome

- Mechanism of Action: IgG auto-Abs against intrinsic Ag of GBM
- Organ specific: Renal GBM
- Signs & Symptoms:
 - Genetic associations: HLA-DR2
 - Adults
 - Rapidly progressive glomerulonephritis (hematuria)
 - Pulmonary hemorrhage (hemoptysis)
 - Preceding URI

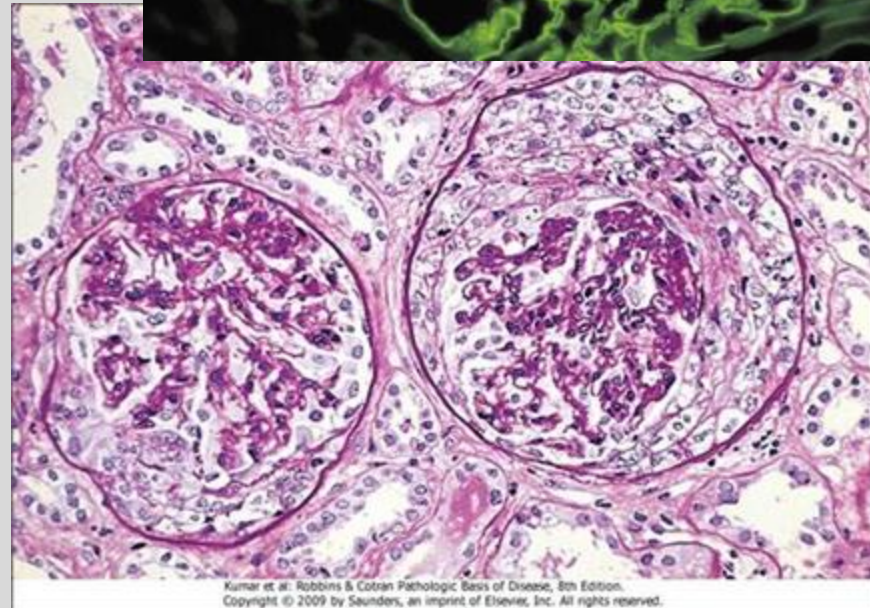
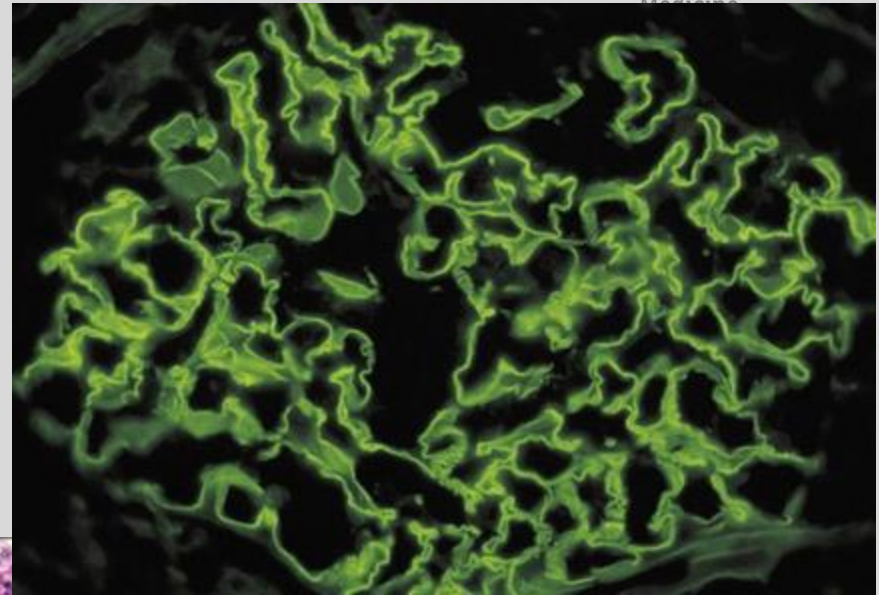
Goodpasture's Syndrome

● Diagnosis:

- Anti-GBM Abs >90%
- CXR: pulmonary infiltrate
- Light microscopy
- Immunofluorescence microscopy: Linear

● Treatment:

- Plasmapheresis
- Cyclophosphamide
- Corticosteroids



Kumar et al: Robbins & Cotran Pathologic Basis of Disease, 8th Edition.
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http://www.stepwards.com/?page_id=1190

<https://microbeonline.com/elisa-test-for-antigenantibody-detection/>

Autoimmune Diseases of the Skin

Pemphigus Vulgaris

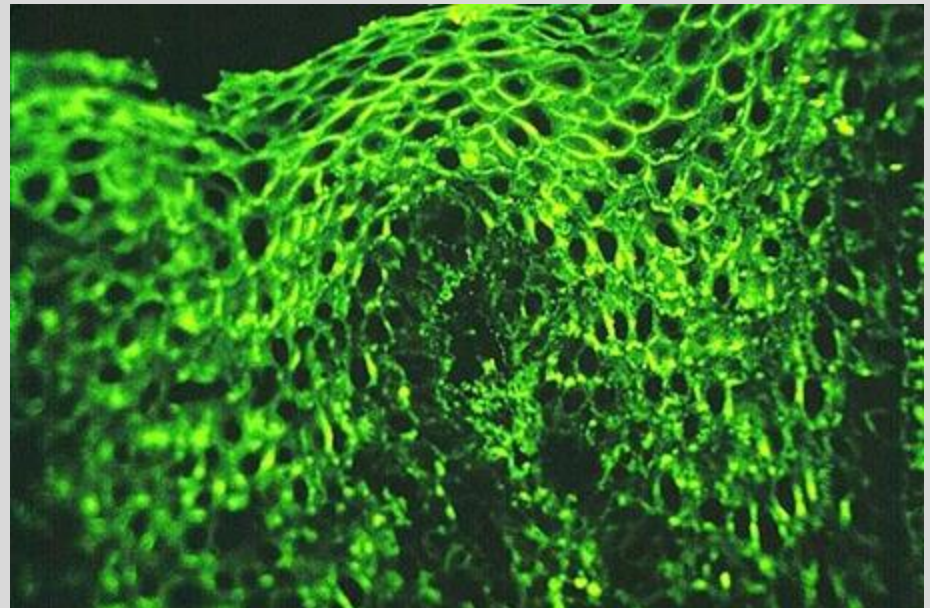
- Mechanism of Action: IgG auto-Abs against desmoglein 1 and/or 3 (desmosome)
- Organ-specific: Mucosal &/or cutaneous epithelial cell junctions
- Signs & Symptoms:
 - Initially, oral mucosa lesions
 - Painful, flaccid bullae
 - Positive Nikolsky sign



Lawrence L., James, W. D., et al: *Andrews' diseases of the skin*, ed 12, Philadelphia, 2016, Elsevier.

Pemphigus Vulgaris

- Diagnosis:
 - Anti-desmoglein 1/3 Abs
 - Light microscopy: Acantholysis
 - Immunofluorescence: Reticular
- Treatment:
 - Corticosteroids
 - Rituximab

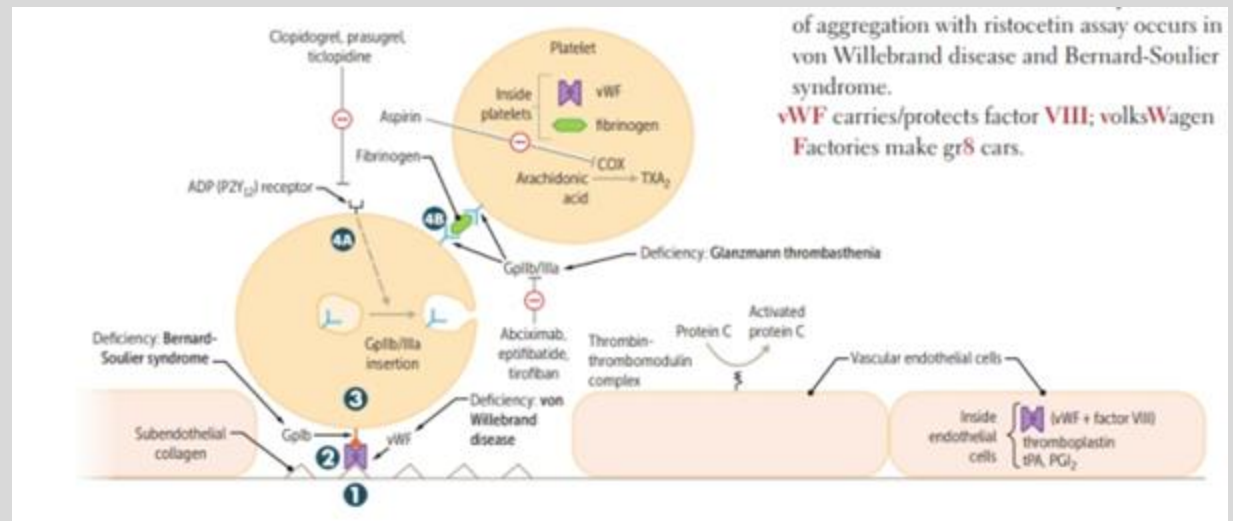


<https://www.pathologystudent.com/whats-the-difference-between-pemphigus-vulgaris-and-bullous-pemphigoid/>

Autoimmune Diseases of the Vascular System

Immune Thrombocytopenic Purpura

- Mechanism of Action: Auto-Abs against GpIIb/IIIa
- Organ specific: Blood, platelets
- Signs & Symptoms:
 - Purpura, mucocutaneous bleeding, splenomegaly
 - Pallor, fatigue

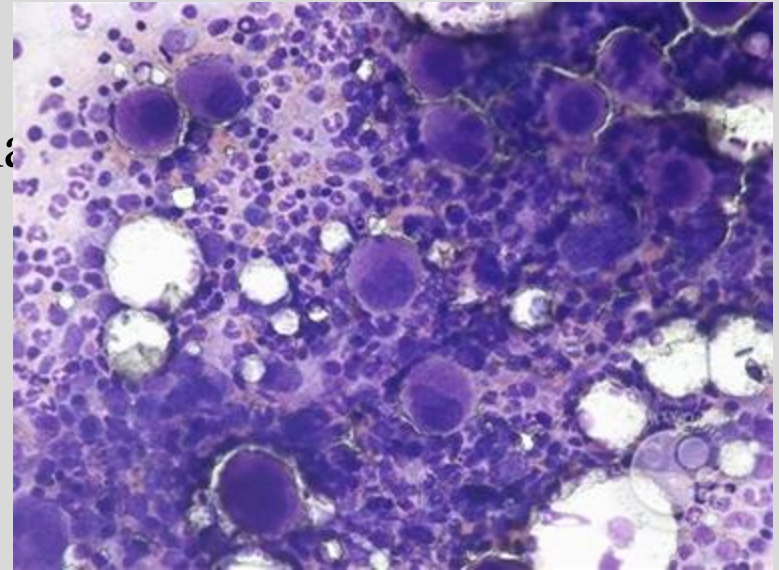


First Aid for USMLE Step 1 2019

Immune Thrombocytopenic Purpura

- Diagnosis:

- CBC: Isolated thrombocytopenia
- Diagnosis of exclusion
- Blood smear
- Bone marrow biopsy (adults)



<https://imagebank.hematology.org/image/16992/bone-marrow-ityp>

- Treatment:

- Children: none
- Adults: Corticosteroids, rituximab, IVIG, & anti-IgD; platelets, splenectomy
- D/C drugs inhibiting platelet function

Platelets below 50,000

Antiphospholipid Syndrome

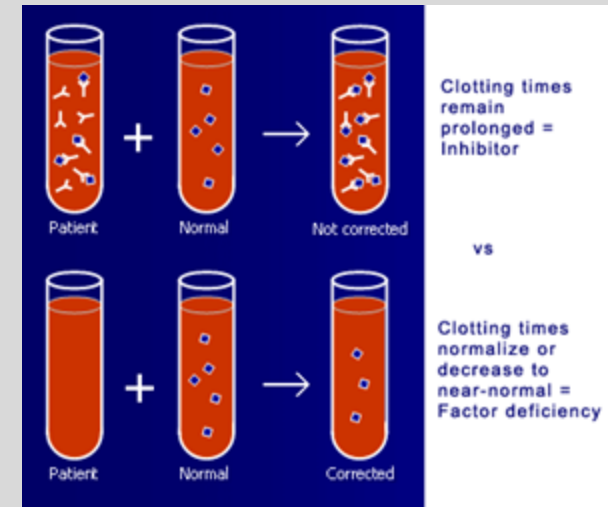
- Mechanism of Action: Auto-Abs vs. proteins that bind phospholipids → thrombin (acquired hypercoagulability; 1' or 2'')
- Lupus anticoagulant
- Cardiolipin (IgG or IgM)
 - β_2 -glycoprotein I (IgG or IgM)
- Non-organ specific: Hematologic; catastrophic APS ≥ 3 organ systems
- Signs & Symptoms:
 - Recurrent thrombosis: DVT, PE, CVA, MI, etc.
 - Spontaneous abortions
 - Livedo reticularis



Antiphospholipid Syndrome

● Diagnosis:

- Sapporo APS Classification Criteria
 - Labs, ≥ 1 : + Anticardiolipin (IgG and/or IgM), + Lupus Anticoagulant, + β_2 -glycoprotein I
 - Clinical ≥ 1 : Vascular thrombosis, pregnancy morbidity



- Treatment: lifelong anticoagulation with warfarin (heparin if pregnant)

Summary Slide

Diseases	AB	CLINICAL SIGNS
Hashimoto's Thyroiditis		
T1DM		
Sjogren's Syndrome		
PBC		
Pernicious Anemia		
UC		
PSGN		
Goodpasture's Syndrome		
Pemphigus Vulgaris		
ITP		
Antiphospholipid Syndrome		

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>

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